



Faculty of Engineering
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CMPS451: Big Data
Project Proposal: Smart Music Recommendation System Based on
User's Choices

Submitted by:

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A) Idea

Due to the huge amount of music available online, users find it challenging to find music tracks that are similar to a particular track that they enjoy listening to.

The objective of this project is to develop a music recommendation engine that creates playlists of music tracks like the one chosen by the user. This is achieved by analysing the features of the selected music tracks and recommending those that are similar to them.

The benefits of this proposal include improving the user experience.

B) Datasets

<https://www.kaggle.com/datasets/tomigelo/spotify-audio-features>

<https://www.kaggle.com/datasets/mrmorj/dataset-of-songs-in-spotify>

<https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset>

<https://www.kaggle.com/datasets/zaheenhamidani/ultimate-spotify-tracks-db>

<https://www.kaggle.com/datasets/tonygordonjr/spotify-dataset-2023>

<http://millionsongdataset.com/>

C) Planned approach

1. **Data Preprocessing (using Apache Spark):**

Load and clean the dataset, select relevant audio features, and normalize numerical values.

2. **Exploration & Visualization:**

Analyse feature distributions and relationships to better understand the data.

3. **Similarity-Based Recommendation:**

Represent each song as a feature vector and compute similarity using **cosine similarity**. A similarity matrix will be precomputed for efficiency.

4. **Playlist Generation:**

Given an input song, the system retrieves the top N most similar songs and generates a playlist.

5. **Evaluation:**

Evaluate the quality of recommendations and compare results with alternative methods (e.g., KNN) if needed.